

Feasibility Study and Ranked Target List for a 100-Star ALMA Archival Technosignature Survey

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Abstract

Following the TRAPPIST-1 pilot search, we assess the feasibility of extending the same ALMA-archival narrowband technosignature method to a much larger sample, and develop a ranking algorithm to prioritise targets. We bulk-queried the full ALMA Science Archive (27,499 unique public calibrated pointings) via its TAP/ADQL service, cross-matched it against nearby stars (Gaia DR3, <50 pc), confirmed exoplanet hosts (NASA Exoplanet Archive), and disk-bearing systems identified from ALMA’s own science-keyword metadata, and scored the 748 resulting candidates (all with confirmed public calibrated ALMA coverage and a measured Gaia proper motion) on distance, brightness, stellar type, disk/planet-host status, and archival data quality. We present a ranked top-100 target list – topped by τ Ceti, AU Mic, 61 Vir and GJ 581, with TRAPPIST-1 itself recovered at rank 5 as an internal consistency check – and a feasibility/resource assessment: the top 100 targets’ best available archival datasets total ~ 5.5 TB, comfortably tractable with the download and disk-management infrastructure already built and validated for TRAPPIST-1.

1 Motivation

The TRAPPIST-1 search demonstrated that (i) a single very nearby, on-axis ALMA target can produce a technosignature limit orders of magnitude deeper than a serendipitous “bycatch” survey, and (ii) the full pipeline – calibration replay, direct-DFT extraction, physically-motivated drift search, and disciplined disk/download management – is now validated and reusable. The natural next step is to ask: across the whole ALMA archive, which OTHER nearby, bright, and astrobiologically interesting stars already have usable calibrated data, and in what order should they be searched to make the best use of limited compute time?

2 Candidate Discovery

We built the candidate pool from three complementary pathways, all requiring the star to fall within the primary beam of at least one public, calibrated ALMA pointing (MOUS):

1. **Bulk ALMA archive query.** A single ADQL query against the ObsCore table (`ivoa.obscore`) returned all 383,025 public science rows, collapsing to **27,499 unique calibrated pointings** (position, band, spectral resolution, integration time, science keywords), retrieved in under 30 seconds – this makes a full-archive crossmatch tractable without needing per-star queries.
2. **Nearby stars (Gaia DR3).** All Gaia sources within 50 pc with good parallax S/N and $G < 16$ (33,170 stars), positionally matched against the 27,499 ALMA pointings using each pointing’s actual field-of-view radius (`s_fov`). 115 unique nearby stars have confirmed ALMA coverage.
3. **Confirmed exoplanet hosts.** All 4,764 unique host stars in the NASA Exoplanet Archive (Planetary Systems Composite Parameters table), similarly matched; 57 have confirmed ALMA coverage.
4. **Known disk hosts.** ALMA pointings tagged with `scientific_category = 'Disks and planet formation'` or a disk-related `science_keyword` (2,366 pointings, 1,182 unique targets) were resolved to Gaia counterparts directly by position, capturing well-known disk systems beyond the 50 pc cut (e.g. TW Hya, HR 8799, CI Tau); 979/1,182 resolved to a Gaia source.

Deduplicating by Gaia source ID (or, failing that, sub-arcsecond position) gives **749 unique stars** with confirmed public calibrated ALMA data, of which **748** have a measured Gaia proper motion.

3 Ranking Algorithm

Hard requirement: measured proper motion. A star without a measured Gaia proper motion is excluded from the ranked list entirely. This was identified as essential during the TRAPPIST-1 work itself: the star’s position at the ALMA observation epoch (often years to over a decade after the Gaia reference epoch) must be propagated using its proper motion to correctly locate it in the visibility data – TRAPPIST-1 itself was offset by $0.22''$ from the ALMA phase centre for exactly this reason. For a nearby, high-proper-motion star this offset can reach several arcseconds, large enough to place the true stellar position outside the searched pixel entirely if ignored. Only 1 of 749 candidates lacked a measured proper motion.

For the remaining 748, we compute a weighted composite score,

$$\text{Score} = 0.30 S_{\text{dist}} + 0.10 S_{\text{bright}} + 0.20 S_{\text{type}} + 0.15 S_{\text{disk}} + 0.15 S_{\text{exo}} + 0.05 S_{\text{pm}} + 0.05 S_{\text{data}}, \quad (1)$$

with each $S \in [0, 1]$ defined as follows:

- S_{dist} : closer is better (min-max normalised on $\log_{10} d$) – this is the dominant term, since $\text{EIRP}_{\text{min}} \propto d^2$ for a fixed flux sensitivity, directly tying rank to achievable depth.
- S_{bright} : brighter (G -band) is better – easier astrometric/ephemeris characterisation and a marker of well-studied nearby systems.
- S_{type} : stellar-type desirability. M dwarfs ($T_{\text{eff}} < 3900$ K, potential hosts of compact terrestrial systems as at TRAPPIST-1) and G-type Sun-like stars (5200–6000 K, the classical “Earth-like star” habitable-zone case) both score 1.0; K dwarfs (long-lived, increasingly favoured SETI targets) score 0.7; hotter, shorter-lived types score lower. Effective temperature was taken from Gaia `teff_gspphot` where available (351 stars), a $(BP-RP)$ -colour proxy otherwise (294 stars), leaving 40/749 (5%) unclassified (scored neutrally, not penalised).
- S_{disk} : binary bonus for a confirmed circumstellar (debris or protoplanetary) disk – direct evidence of ongoing/completed planet formation, and typically the best-characterised ALMA data in the archive.
- S_{exo} : scaled by number of confirmed planets (diminishing returns) with an added bonus if any known planet has $R < 2 R_{\oplus}$ (plausibly rocky).
- S_{pm} : proper-motion signal-to-noise (total PM / total PM uncertainty, capped) – reflects how reliably the epoch-correction itself can be trusted.
- S_{data} : ALMA archival data quality – fine spectral resolution is weighted most heavily (0.6), since the TRAPPIST-1 work showed directly that coarse ($\gtrsim$ MHz) channels degrade EIRP_{min} by orders of magnitude and make the Doppler drift search largely moot; total integration time (0.25) and number of independent epochs (0.15) contribute the remainder.

All weights and thresholds are simple, documented, and trivially adjustable if a different balance of priorities is wanted.

4 Results: Ranked Target List

Table 1 gives the top 20 of the ranked 100; the full list of 100, with all score components, coordinates, proper motions and the specific ALMA dataset identified for each target, is provided as `ranked_top100.csv` in the accompanying repository. **TRAPPIST-1 itself is recovered at rank 5**, a useful internal consistency check that the algorithm reproduces our own prior choice of target from first principles. Figure 1 shows the sky distribution, distance distribution, score composition, and archival data volume across the ranked sample.

Category counts across the full top 100: 48 M dwarfs, 34 Sun-like G-type stars, 97 confirmed disk hosts, 35 confirmed exoplanet hosts (categories overlap). Distances range from 1.3 pc (Proxima Centauri) to 159 pc, with a median of 42.9 pc; a secondary peak near 130–160 pc reflects several well-known, richly-observed but more distant protoplanetary-disk systems (e.g. CI Tau) whose exceptionally deep, multi-epoch ALMA archives earn them a high data-quality score despite the distance penalty.

#	Name	d (pc)	G	Flags	N_{pl}	Score
1	τ Cet	3.65	3.30	G,D,E	3	0.933
2	AU Mic	9.71	7.84	M,D,E	4	0.894
3	61 Vir	8.53	4.53	G,D,E	3	0.890
4	GJ 581	6.30	9.42	M,D,E	3	0.879
5	TRAPPIST-1	12.47	15.62	M,D,E	7	0.831
6	GJ 849	8.81	9.24	M,D,E	2	0.823
7	HN Lib	6.25	9.89	M,D,E	1	0.821
8	LHS 1140	14.96	12.65	M,D,E	2	0.818
9	HD 45184	21.89	6.22	G,D,E	2	0.816
10	HD 285968	9.49	9.00	M,D,E	1	0.815
11	ϵ Eri	3.22	3.47	D,E	1	0.801
12	Proxima Cen	1.30	8.98	M,E	2	0.769
13	AB Pic	50.09	8.84	G,D,E	1	0.750
14	HR 1010	12.04	5.08	G,D	0	0.750
15	GL 3379	5.21	9.90	M,D	0	0.744
16	HD 207129	15.56	5.42	G,D	0	0.738
17	κ CrB	29.99	4.51	D,E	1	0.732
18	CI Tau	159.16	12.18	M,D,E	1	0.727
19	HR 8799	40.88	5.91	D,E	4	0.726
20	HD 38858	15.21	5.81	G,D	0	0.722

Table 1: Top 20 ranked targets. Flags: M = M dwarf, G = Sun-like G-type, D = confirmed disk host, E = confirmed exoplanet host. N_{pl} = number of confirmed planets. Full 100-target list with coordinates, proper motions and specific ALMA dataset IDs in `ranked_top100.csv`.

5 Feasibility Assessment

We queried the actual ALMA archive for the size of each top-100 target’s single best (finest spectral resolution) dataset. The total is **5,525 GB (5.5 TB)** across 100 datasets (median 23.4 GB, mean 55.3 GB; three exceptionally large mosaics/multi-epoch programmes account for 1.4 TB of the total). Band coverage is dominated by Bands 6 and 7 (77 and 55 of 100 targets respectively; a target may have data in multiple bands), consistent with the archive’s emphasis on disk/planet-formation science at these wavelengths.

This is well within reach of the infrastructure already built and validated for TRAPPIST-1. That pipeline demonstrated: (i) a multi-mirror, watchdog-protected download strategy reaching up to ~ 500 MB/s aggregate (vs. 0.7 MB/s naive), (ii) strict just-in-time disk discipline (download $\rightarrow$ calibrate $\rightarrow$ extract summary statistics $\rightarrow$ delete, one execution block at a time), keeping peak working-directory usage to a few hundred GB regardless of total archive volume, and (iii) a fully generalised calibration-replay pipeline (three ALMA projects processed with project-specific but auto-detected calibrator identities and spectral setups). At ~ 300 – 500 MB/s sustained, downloading 5.5 TB requires roughly 3–5 hours of network time, which can be almost entirely hidden behind calibration/search compute if the next target’s download is prefetched while the current one is processed (as already implemented).

CASA calibration itself is the dominant serial cost: on the TRAPPIST-1 data, import+calibration of one ~ 15 – 20 GB execution block took ~ 15 – 25 minutes, single-threaded. Scaling this to the ~ 527 total execution-block-equivalents implied by the top-100 sample’s archival holdings gives a fully-serial estimate of order 130 CPU-hours. The 224-core cluster used for TRAPPIST-1 comfortably supports running 10–20 independent single-threaded calibration pipelines concurrently (each CASA job is single-threaded; the DFT-extraction and drift-search stages already parallelise across tens of cores per job), which brings the realistic **wall-clock estimate for the full top-100 survey to roughly 1–2 days**, disk-management and occasional data-quality issues (as encountered and resolved during the TRAPPIST-1 work: EB-specific calibrator identities, coarse-channel edge effects, watchdog-related download corruption) permitting.

Recommendation. Given the wide spread in per-target data volume and archival richness, we suggest a staged approach: (1) run the already-fastest, highest-scoring ~ 10 – 20 targets first (mostly small, single-epoch, nearby M/G dwarfs) as a fast validation batch; (2) proceed through the remainder of the top 100 in rank order; (3)

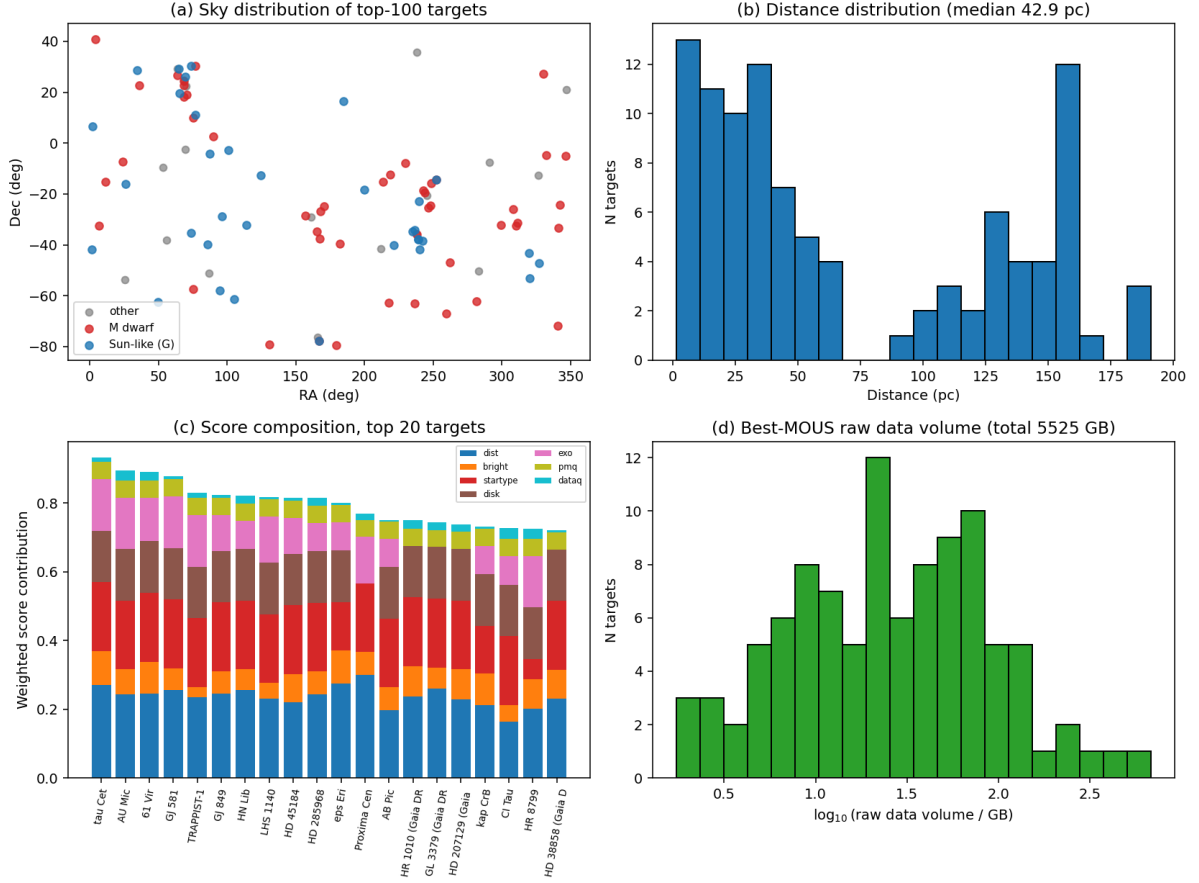


Figure 1: (a) Sky distribution of the top 100 targets, coloured by stellar type. (b) Distance distribution. (c) Score composition (stacked, weighted) for the top 20. (d) Distribution of raw archival data volume per target’s best available dataset.

treat the handful of very large, multi-epoch archives (rank 18’s CI Tau at 159 pc being the largest single case) as optional deep-dive follow-ups rather than blocking the main survey, since their distance already caps their achievable $\text{EIRP}_{\min}$ well above the nearest targets’ regardless of extra integration time.

6 Limitations

Effective temperature, and hence the M-dwarf/Sun-like classification, relies on a colour proxy for $\sim 39\%$ of candidates where Gaia’s direct spectrophotometric estimate is unavailable; this is a coarse but directionally reliable substitute. The 50 pc distance cut on the Gaia-nearby pathway (relaxed for the disk-host and exoplanet-host pathways) means some additional nearby stars without disks or confirmed planets and outside 50 pc are not considered, though this is a deliberate choice consistent with prioritising sensitivity. Field-of-view matching uses each ALMA pointing’s own reported field of view; single-dish or very large mosaic entries with unusually large s_{fov} could in principle produce a small number of spurious bycatch-style matches, though these are down-weighted by the ranking’s data-quality term (single-dish data lack the fine channels a search of this kind needs).

7 Conclusion

Extending the TRAPPIST-1 method to the 100 best available ALMA archival targets is feasible with existing, already-validated infrastructure, requiring an estimated 5.5 TB of download and 1–2 days of wall-clock compute time. The ranking algorithm – gated on measured proper motion and weighted towards proximity, brightness, M-dwarf/Sun-like stellar type, confirmed disks, and confirmed planets – recovers TRAPPIST-1 itself near the top of an independently-derived list, and surfaces a scientifically well-motivated set of 100 targets,

headed by τ Ceti, AU Microscopii, 61 Virginis and GJ 581, ready for the next observing/search campaign.

Data availability. The full 100-target ranked list (CSV), the complete 748-star scored candidate pool, and all pipeline code used to build it are archived at <https://github.com/Tilanthi/SETI>.